

Q1. Explain the role of Tokenization in Natural Language Processing (10 Marks)

Introduction

Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, interpret, and generate human language. Before a machine can process text, the text must be converted into smaller meaningful units. This process is known as Tokenization.

Definition

Tokenization is the process of splitting a text document into smaller units called tokens. Tokens may be sentences, words, characters, or sub-words. It is the first and one of the most important preprocessing steps in NLP.

Need for Tokenization

1. Converts unstructured text into structured form.
2. Helps machines analyze language efficiently.
3. Enables feature extraction for machine learning models.
4. Improves accuracy of text mining and information retrieval systems.

Types of Tokenization

1. Sentence Tokenization – Divides text into sentences.
2. Word Tokenization – Divides sentences into words.
3. Character Tokenization – Splits text into characters.
4. Sub-word Tokenization – Breaks complex words into smaller units.

Working Example

Input: "NLP is transforming the world."

Output Tokens: [NLP] [is] [transforming] [the] [world]

Diagram

Input Text → Tokenizer → Tokens → NLP Model

Applications

- Search Engines
- Chatbots
- Sentiment Analysis
- Machine Translation
- Text Summarization
- Question Answering Systems

Advantages

- Simplifies text processing.
- Improves computational efficiency.
- Forms the basis for advanced NLP tasks.

Challenges

- Handling punctuation and special symbols.
- Multiple meanings of words.
- Language-specific complexities.

Conclusion

Tokenization is a fundamental NLP technique that transforms raw text into meaningful tokens. It serves as the foundation for almost every NLP application and significantly influences the performance of language-processing systems.

Q2. Analyze the differences between Syntax, Semantics, and Pragmatics in Natural Language Understanding (10 Marks)

Introduction

Natural Language Understanding (NLU) is a subfield of NLP concerned with enabling computers to understand human language. Syntax, Semantics, and Pragmatics represent three levels of language understanding.

Syntax

Syntax refers to the grammatical structure of a sentence and the rules governing word arrangement.

Example: "The boy plays football." is syntactically correct.

Semantics

Semantics deals with the meaning conveyed by words, phrases, and sentences.

Example: "The boy plays football." means that a boy is participating in a football game.

Pragmatics

Pragmatics focuses on language interpretation based on context, intention, and real-world knowledge.

Example: "Can you pass the salt?" is pragmatically a request, not merely a question about ability.

Comparison Table

Syntax – Concerned with structure and grammar.

Semantics – Concerned with literal meaning.

Pragmatics – Concerned with contextual meaning and intention.

Examples

Sentence: "I saw a man with a telescope."

Syntax determines sentence structure.

Semantics identifies possible meanings.

Pragmatics uses context to infer who has the telescope.

Importance in NLU

- Syntax helps parse sentences.
- Semantics extracts meaning.
- Pragmatics understands user intent and context.

Applications

- Virtual Assistants
- Chatbots
- Machine Translation
- Information Retrieval Systems
- Question Answering Systems

Conclusion

Effective language understanding requires all three levels. Syntax provides structure, semantics provides meaning, and pragmatics provides contextual interpretation. Together they enable computers to communicate more naturally with humans.

Q3. Illustrate the application of k-Nearest Neighbor (k-NN) algorithm in document classification (10 Marks)

Introduction

The k-Nearest Neighbor (k-NN) algorithm is a supervised machine learning algorithm used for classification and regression. In document classification, it assigns a category to a document based on the categories of similar documents.

Concept

The algorithm assumes that similar documents belong to the same class. Classification is performed by examining the nearest neighbors of a new document.

Working Steps

1. Collect labeled training documents.
2. Convert documents into numerical vectors using techniques such as TF-IDF.
3. Choose a value of k.
4. Compute distances between the new document and training documents.
5. Select the k nearest neighbors.
6. Assign the majority class.

Distance Formula

Euclidean Distance is commonly used to measure similarity between document vectors.

Illustration

Training Data:

- D1 – Sports
- D2 – Sports
- D3 – Politics
- D4 – Politics

New Document: "The team won the championship."

Nearest neighbors belong mainly to Sports, therefore the document is classified as Sports.

Diagram

Training Documents → Feature Extraction → Distance Calculation → k Nearest Neighbors → Classification

Applications

- News Categorization
- Email Spam Detection
- Sentiment Analysis
- Topic Identification
- Recommendation Systems

Advantages

- Simple and intuitive.
- No explicit training phase.
- Effective for small datasets.

Limitations

- High computation cost for large datasets.
- Sensitive to irrelevant features.
- Performance depends on the choice of k.

Conclusion

k-NN is an effective and widely used algorithm for document classification. By comparing a new document with previously labeled documents, it can accurately determine the appropriate category and support many real-world text mining applications.